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assignment 10 - magnetic field, magnetic force

try first without hints.

READ before:

A magnetic field can deflect a moving charge but can not change its speed or KE.

A magnetic field can deflect a moving charge only if there is a component of the velocity perpendicular to B.

Check your equation sheet:

A moving charge with a velocity V inside a magnetic field is acted upon by a force = $qVB\sin(\alpha)$
 α is the angle between V and B .

A wire carrying current I of length L inside a magnetic field is acted upon by a force = $IL\sin(\alpha)$. α is the angle between I and B

$\tau = N I A B \sin \theta$. This is the torque on a current-carrying loop in a uniform magnetic field. This equation can be shown to be valid for a loop of any shape. The loop carries a current I , has N turns, each of area A , and the perpendicular to the loop makes an angle θ with the field B

Quiz Type Graded Quiz

Points 111.5

Assignment Group assignments

Shuffle Answers Yes

Time Limit No Time Limit

Multiple Attempts Yes

Score to Keep Latest

Attempts 2

View Responses	Always
Show Correct Answers	No
One Question at a Time	No
Require Respondus LockDown Browser	No
Required to View Quiz Results	No
Monitor Required	No

Due	For	Available from	Until
Mar 17	Everyone	-	Mar 17 at 11:59pm

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